

CONTACT

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PROFESSIONAL SUMMARY

- Tech Lead with 15+ years experience cloud-native fault-tolerant, high availability solutions. As an expert in Java/Spring and Angular with a focus on creating modular APIs and adaptive UIs, I have proven track record of leading teams and building o-to1 prototypes.
- An engineer that collaborates with AI tools like GitHub Copilot, n8n and GitLab Duo to accelerate development velocity
- A force multiplier with a passion for mentoring junior engineers, sets examples of good code and best practices and helps other ICs contribute to their full potential
- tumuhairwe.github.io

SKILLS

- **Technical:** Docker & Kubernetes, Spring, Kafka, SQL, NoSQL & Databases
- **Modern Frontend/Backend:** Backend (Spring Boot, Java) and Frontend (Typescript, Angular)
- **Certifications:** AWS Developer and AWS Solutions Architect
- **AI & Productivity Tools:** GitHub Copilot, Gemini, n8n, GitLab Duo
- **Architectural Patterns:** API Design for State Management (GraphQL & RESTful APIs), Modular Design.
- **Monitoring:** New Relic, Datadog & Splunk
- **Other:** Tech Debt Management, API Design, Mentoring and coaching

JUDES TUMUHAIRWE

STAFF SOFTWARE ENGINEER | TECH LEAD



EXPERIENCE

Senior Software Engineer BILL - San Jose, CA

08/2024 - Present

Leading the technical evolution for the Invoice Financing (IF) platform, a high-stakes financial ecosystem generating \$2.4M – \$2.6M in net new quarterly revenue

Impact: The IF teams brings in about 2.4 to 2.6 million dollars every quarter in net new revenue.

My Role

■ **Strategic Monolith Modernization:** Spearheading a multi-year architectural roadmap to decompose a 15-year old revenue-critical monolith into a high-throughput microservices architecture. This initiative directly parallels large-scale redesigns of global invoicing systems to streamline data contracts and evolve legacy data models.

■ **AI-Augmented Risk Infrastructure:** Building the Eligibility Checker for the Identity Verification and Risk Assessment engine. Leveraged AI tools to rapidly prototype the logic for user eligibility, accelerating the development of the short-term collateral borrowing lifecycle while ensuring strict data protection and manual verification of AI-generated code for financial accuracy.

■ **Risk Assessment & Identity Lifecycle:** Building features e.g. Eligibility Checker for the Identity Verification and Risk Assessment engine that serves as the critical gatekeeper for the short-term collateral borrowing lifecycle. This system automates user eligibility checks for immediate cash advances, ensuring secure handling of sensitive financial PII and adherence to strict data protection standards

■ **Scalable Transactional Infrastructure:** Designed and implemented the "Pay Me Now" transactional engine and automated communication pipelines using SendGrid. Used AI to synthesize and distill themes from historical transaction failure patterns, allowing for the proactive design of more resilient, privacy-compliant features.

■ **Technical Design Authority & Productivity:** I create comprehensive design docs and lead technical reviews to ensure reliability during the transition from Micronaut to distributed microservices. I use AI to automate repeatable tasks such as drafting docs and generating flow charts and sequence diagrams using Lucid AI, significantly increasing team velocity and developer productivity.

Environment

- Micronaut, Gradle, Oracle, GraphQL, SonarQube, Docker, Kafka, Redis, AWS Splunk

Tech Lead Capital One - McLean, VA

07/2022 - 11/2023

As the Tech Lead for the Internal Digital Recoveries(IDR) teams in the Card Tech org I was responsible for leading 3 different vendor teams to bring recoveries in-house (instead of being sold to debt-buyers).

Impact: The IDR project lead to \$208 million in net new revenue.

My Role:

■ **Financial Compliance & Audit Readiness:** Engineered mission-critical services under strict SOX and financial regulatory regimes, implementing data governance standards that ensured 100% auditability across global markets (US, EU, EMEA).

■ **High-Stakes Technical Governance:** Led technical design sessions and defended architectural decisions before the Design Review Board, ensuring that all recovery-system intents adhered to Privacy by Design principles and enterprise-wide financial standards.

■ **Organizational Leverage:** Facilitated alignment between cross-functional teams (Card Tech, Legal, and Finance) to bring debt recovery in-house, replacing external vendor systems with a centralized, scalable platform

■ **Distributed Systems Leadership:** Led 3 distributed engineering teams in the design, deployment, and maintenance of fault-tolerant microservices using Spring Boot, Kafka, and AWS (RDS, Lambda)

■ **Observability & Resilience:** Followed standardized production support protocols and monitored dashboards using New Relic and Datadog, enabling real-time performance tracking and rapid incident resolution for high-throughput financial transactions.

Environment

Spring Boot, Gradle, Postgres, Angular, SonarQube, Datadog, Docker, Kafka, MongoDB, SAFe, AWS (EC2, RDS, Lambda, CloudWatch), New Relic, Splunk, Snowflake, Java 21

Senior Software Engineer Ford Motor Company - Dearborn, MI

10/2019 - 04/2022

■ My team in architected a high-throughput data operating system for vehicle telemetry. The Command Service functioned as the gatekeeper for millions of global vehicle events. I was in charge of telemetry and event processing for Connected Vehicles. We rapidly prototyped and deployed cloud-native microservices to accelerate internal engineering velocity. If you're looking for 0-to-1 prototyping, I'm your guy.

Impact: The CVDOS project reduced time from ideation to prototype by 65% for engineers at Ford

My Role:

■ Developed high-performance Java/Spring Boot services and Kafka-based streaming pipelines to ingest, validate, and process massive telemetry datasets with 100% accuracy.

■ Mentored and provided technical guidance to junior engineers & created onboarding docs and led blameless post-incident reviews

■ Added features to the Angular/Typescript front-end and mentored junior team members around best practices. Highlighted the importance of non-functional requirements when reviewing code.

■ Fostered a culture of excellence, inclusion, collaboration, and performed code reviews to ensure quality standards were met.

Environment

Agile, Java 17, Spring Boot, SQL Server, Angular, Splunk,

Maven, SonarQube, Redis, Pivotal Cloud Foundry, Google Cloud Platform, PostMan, Git

Software Engineer Cengage Learning - Farmington Hills, MI
04/2017 - 09/2019

My team was responsible for building the subscription platform ("Cengage Unlimited") to counter dwindling textbook sales by making thousands of books at the "Netflix of publishing" available in various formats. Cengage Unlimited is a consumer-facing product used in hundreds of universities all over the US.

Impact: Cengage Unlimited introduced a subscription-based model to publishing which directly increased recurring revenue.

My Role:

- On the Student Dashboard team, I developed core Identity and Access Management (IAM) features for the 'Netflix of Publishing,' managing secure SSO integrations and member authentication for thousands of digital assets while prioritizing consumer privacy.
- Worked with UX/UI designers to implement the student dashboard containing various forms of content (e.g. ebooks, PDFs, video audio and images).
- Led team retrospectives to identify and resolve impediments/blockers.
- Mentored junior software engineers by ensuring adoption of Agile best practices including regular retrospectives, post mortems, encouraging unit & integration tests, etc.
- Coordinated prioritization of bugs and scheduling of production releases with product owners, UX/UI and QA teams

Environment

Agile, Java 8, Spring Boot, RxJava, Hystrix, Maven, Bower, AngularJS, Splunk, Jenkins

Software Engineer Advaita Bioinformatics - Plymouth, MI
02/2016 - 04/2017

My team was responsible for building the tools used to scrape data from public repositories, parse it and upload it into the search index.

Outcome: We created the 1st one-stop shop for genomic data in the bioinformatics industry

- I developed cloud native microservices, writing scripts to collect, parse and upload data from multiple public repositories and improving the user-experience.
- I developed and integrated UI controls with multiple java microservices.
- I wrote bash and perl scripts to download, parse and merge data from multiple public repositories and integrate it into our search engine index and other applications.

Environment

Agile, Spring Boot, Hibernate, Zuul, Eureka, S3, SNS, SQS, RDS, Maven, Grunt, Bower, AngularJS

Software Engineer Ithaka/JSTOR - Ann Arbor, MI
09/2014 - 12/2015

My team ("Content Ingest") was responsible for receiving content (zip files) from publishers of books and journals and itemizing, characterizing, and preparing them for use by other teams (e.g. search-index team) and to build the UI publishers

use to upload and track new journals/issues as they move through different states

Impact: With Content Ingest publishers can quickly upload new editions of journals and make them searchable by all universities, which increased revenue for both Ithaka and the publishers

- I was responsible for deploying jars, starting and monitoring clusters, and running the jars on Hadoop clusters to run transforms on large datasets. I implemented microservices deployed as REST APIs to access metadata about content.
- Proposed and implemented code and architectural changes to move away from Hadoop to REST-apis. Enabled service-location using Eureka.
- Implemented tracking of content as it went through different phases using finite state machine
- Participated in PagerDuty rotation to help customers understand issues, triage and often resolve them.

Environment

Agile, Spring, Cassandra, Hadoop, Swagger, Amazon S3, SNS, SQS, Apache Solr

Software Engineer Domino's Pizza - Ann Arbor, MI

08/2012 - 09/2014

My team ("Platform team") built and scaled the global ordering platform backend, ensuring 99.99% availability for a multi-client ecosystem (Web, iOS, Android) of consumer-facing products during superbowl.

Impact: Scaled the Platform so it could support more front-ends, leading to more orders and increased revenue e.g. serving millions of concurrent users during Superbowl

My Role

- **Extreme Scale & High Availability:** I performance-tuned the core platform to support millions of concurrent users during the Superbowl, managing massive traffic spikes through optimized caching and distributed resource management.
- **Global Architecture & Internationalization:** I spearheaded the internationalization of the backend infrastructure, re-designing data storage and user identity schemas to support rapid market expansion while adhering to diverse regional data residency and localized privacy laws.
- **Latency Optimization & Resilience:** Implemented multi-level caching strategies using EhCache and utilized Gatling/JMeter for rigorous load testing to identify and eliminate system bottlenecks, ensuring a low-latency, personalized ordering experience.
- **System Decoupling & Orchestration:** Decomposed monolithic ordering logic into resilient microservices and implemented Mule Enterprise Service Bus (ESB) for sophisticated service orchestration, message routing, and data transformation.
- **Fault-Tolerant Design:** We built a localized menu and pricing modules to ensure system reliability across multiple locales, maintaining a consistent high-availability posture during complex global rollouts.

Environment

Agile, Spring, JMeter, Jenkins, SoapUI, Oracle, Mule ESB, Cucumber, Gatling, MuleSoft

Software Engineer Ford Motor Company - Dearborn, MI
01/2008 - 07/2012

■ Project 1:

- Part Library 2 is a repository of all the parts used on all the vehicle-programs in Ford PL 2 eased recalls and has saved the company \$7 million annually data integrity gave the designers and business a single view of the data

My Role

- Managed the 'journey' of millions of data records through custom ETL processes, ensuring data integrity and building robust ingestion gatekeepers for high-volume upstream sources.
- Designed and built batch jobs to periodically fetch data from upstream external systems
- Interacted with vendor APIs to extract data, transform it based on dynamic logic, and load (or ignore) it into the new system. Also wrote the UI for users to request ad-hoc batch jobs
- Designed a notification system (email, pager) to be invoked when certain data becomes available

Project 2

Automated Certification System (ACS) is a system to automate the certification of new engines by the EPA. Before, tests and certification took a year but with ACS, it now takes 3 months enabling faster turnaround, improvement in failing designs and faster time to market.

■ My Role

Modernized core data infrastructure by migrating legacy systems to high-efficiency Java services designed for automated, large-scale data processing and validation.

Project 3:

- Cost Reduction Idea Database (CRID2) is a system to automate the collection of ideas from engineers, suppliers, business managers, and partners on how to reduce costs anywhere in the production chain (from conceptualization, design, production to advertising and marketing) across all vehicle programs structured way that can be quickly evaluated, rated, approved and applied.

■ Project 3:

- Implemented a finite state machine to process business logic as data goes from submitted to approved/integrated for various models
- Designed the data model in concert with the data architect

Environment

Agile, J2EE, IBM Websphere, Oracle 10g®, MS SQL Server, IBM Message Queue®, IBM RSA®



EDUCATION

Master of Science (M.S.): Applied Information Systems
Towson University - Towson, MD, US, 05/2006

Bachelor of Science (B.S.): Computer Information Systems
Towson University - Towson, MD, US, 05/2004

Bachelor of Science (B.S.): Business Administration
Towson University - Towson, MD, US, 05/2004



CERTIFICATIONS

AWS Certified Solutions Architect
AWS Certified Developer



PUBLICATIONS

Cloud computing is not just for Fortune 500, 06/02/2023,
Judes Tumuhairwe,
<https://minoritybusinessreview.com/cloud-computing-is-not-just-for-fortune-500>



OPEN SOURCE LEADERSHIP

- **Experiment4J**: As a committer on Experiment4J--java implementation of GitHub Scientist (a popular ruby library for refactoring critical paths)--I create shared abstractions around code refactoring that lower the barrier to entry for devs seeking to refactor monoliths. I use AI to synthesize information and create the roadmap for a safe A/B testing framework for refactoring critical code paths by running experiments, publishing the results to InfluxDB, and defining the entry and exit criteria and the duration of the experiments. Ask me about creating shared abstractions that help devs add features to legacy monoliths.

<https://github.com/tumuhairwe/Experiment4J>

- **MyAdvisor**: A system to help academic advisors and advisees create a plan that satisfies program requirements, organization/department requirements, institution requirements (GenEd reqs) and course requirements to help students graduate in their declared major in the most efficient way. I built this with Spring (back end) and Angular (front end) with Aurora DB

- Active on several mailing-lists including Mule ESB, JBoss Web Services, Netbeans, Apache Axis, Open JPA, etc.



DATABASES

Oracle, MySQL, SQL Server, Postgres
Snowflake, Cassandra
Redshift, DynamoDB, Redis, MongoDB